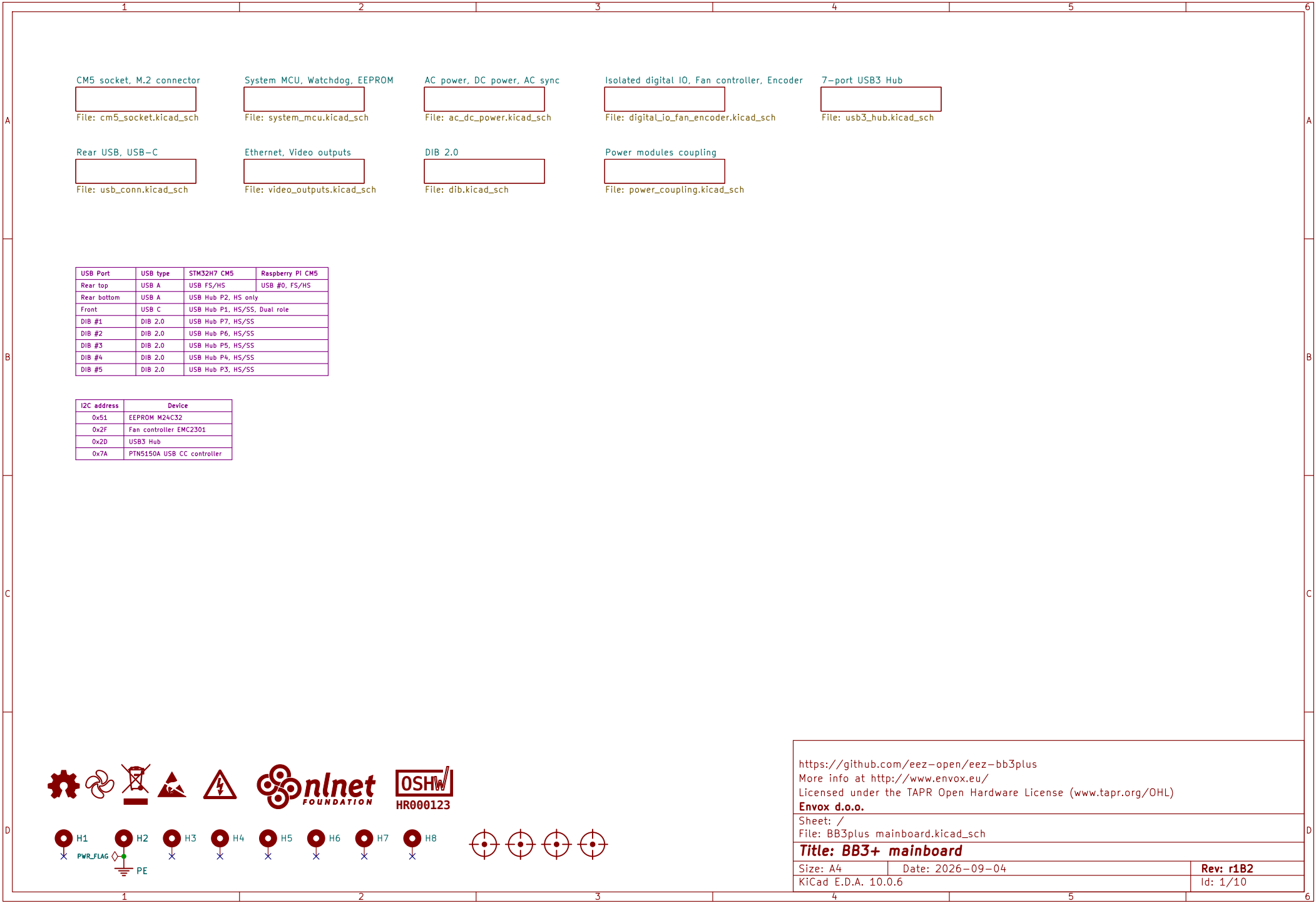
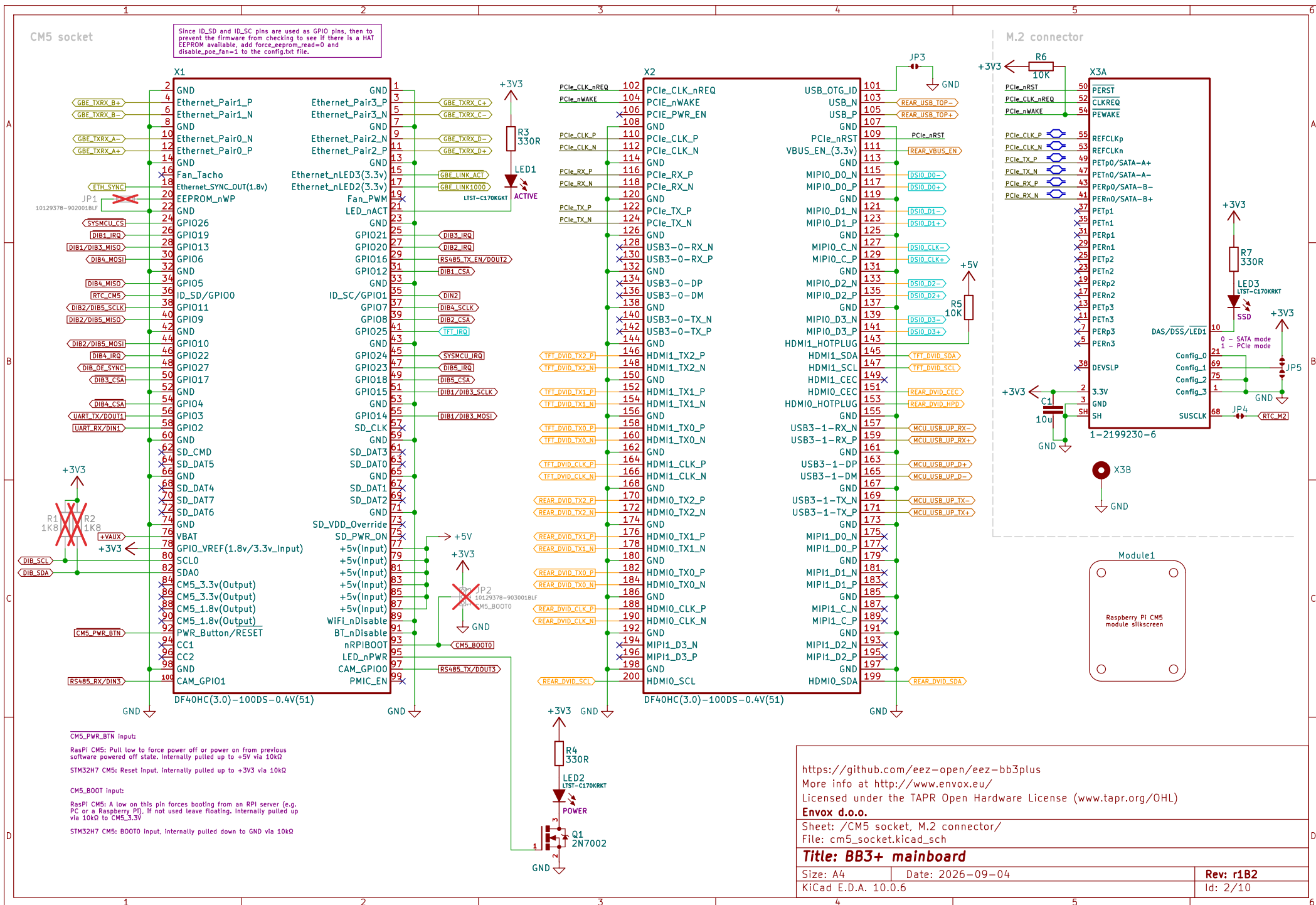






<https://github.com/eez-open/eez-bb3plus>

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Sheet: /CM5 socket, M.2 connector/

File: cm5_socket.kicad_sch

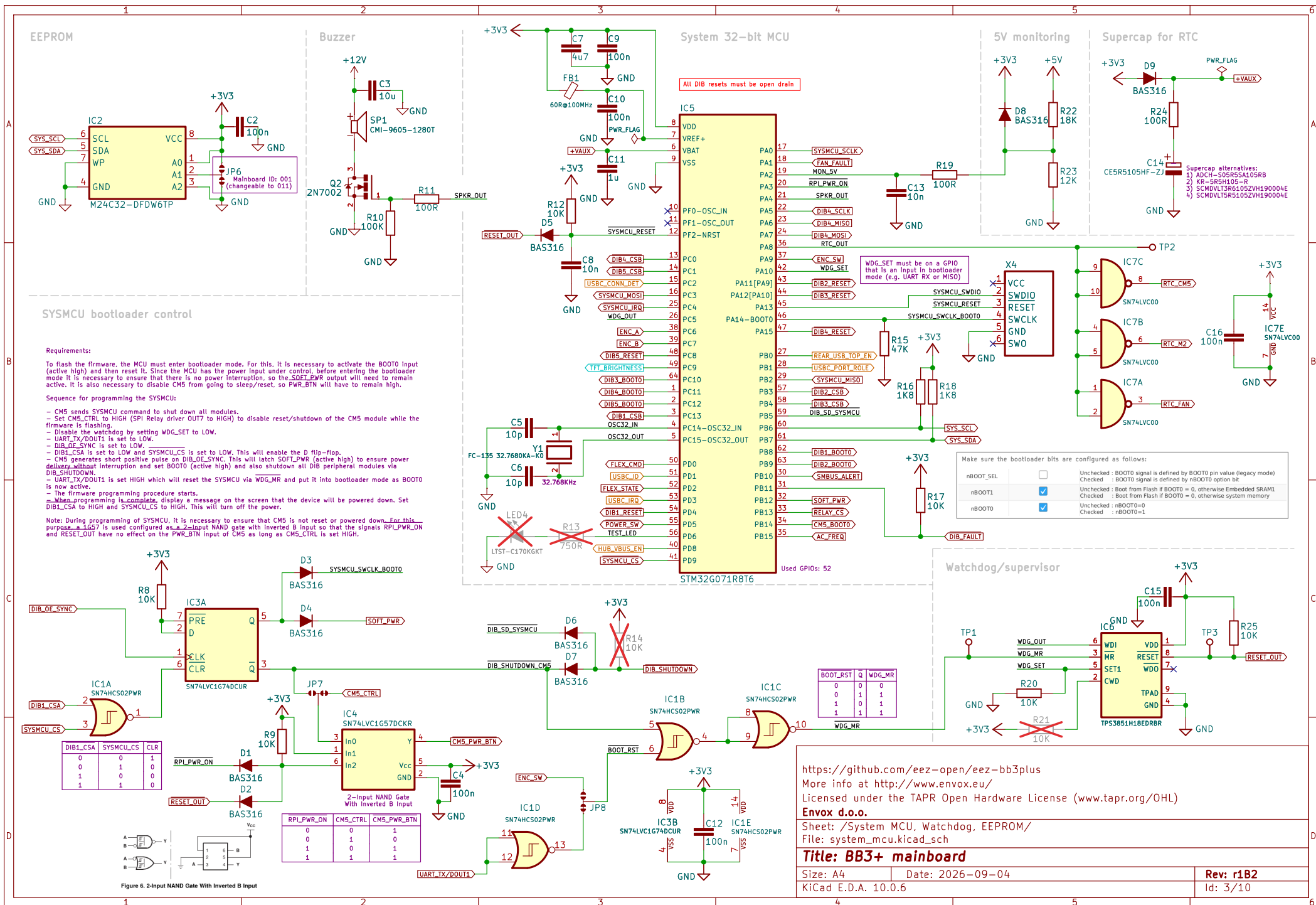
Title: BB3+ mainboard

Size: A4 Date: 2026-09-04

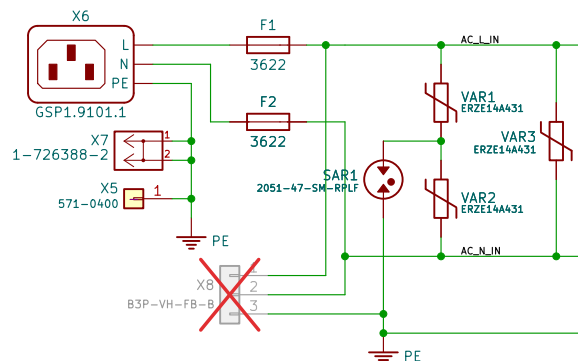
KiCad E.D.A. 10.0.6

Rev: r182

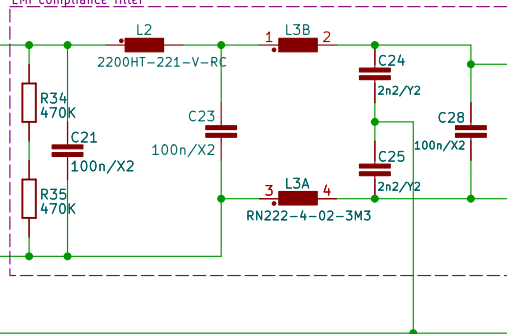
Id: 2/10



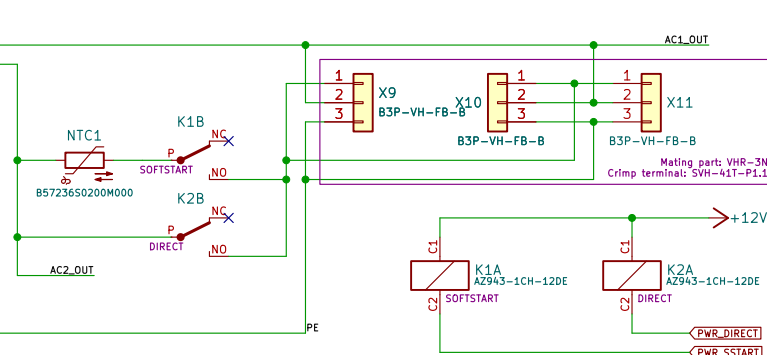
AC inlet, filtering and protection



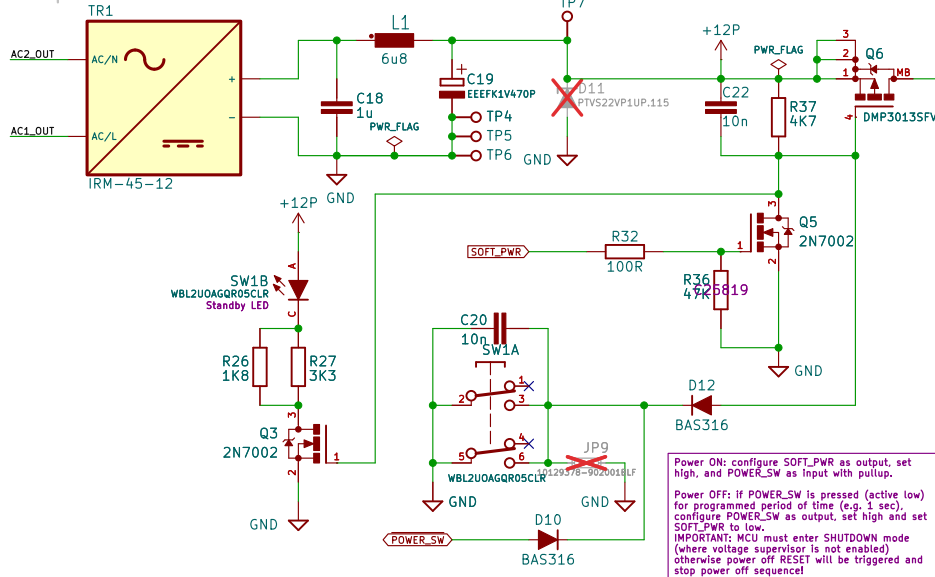
EMI compliance filter



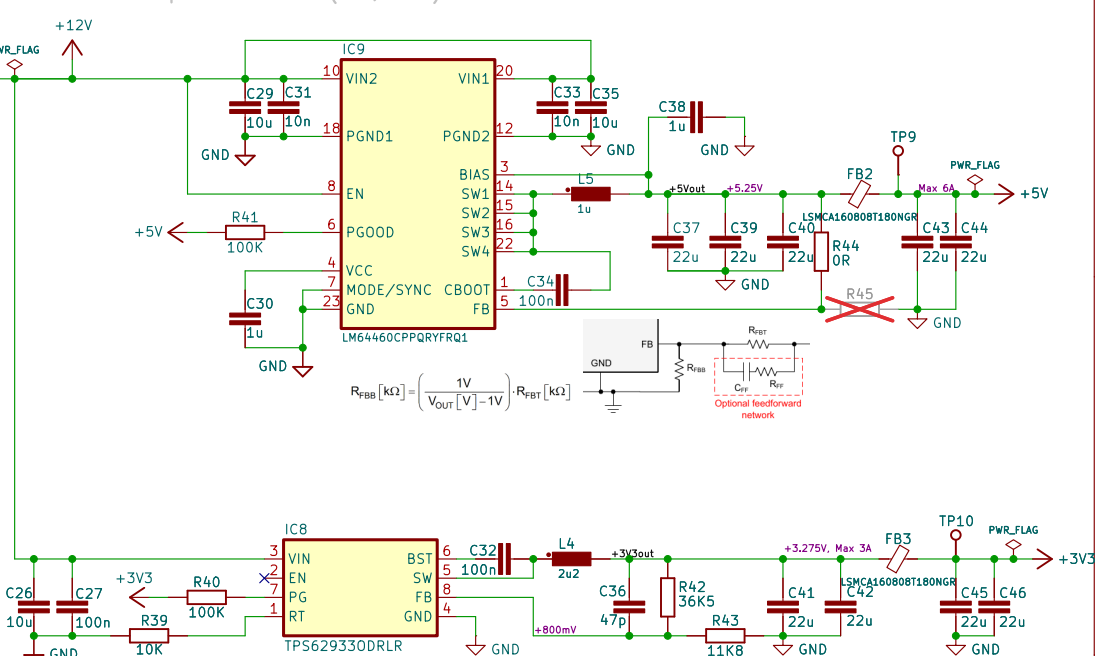
AC power control, Inrush current limiter



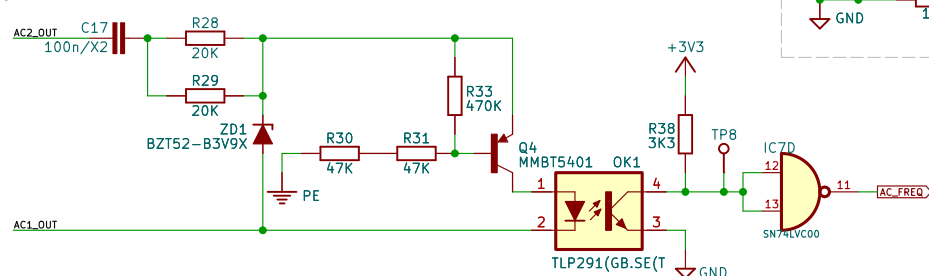
Soft power switch



DC-DC stepdown converters (+5V, +3V3)



AC Mains frequency and PE fault detection



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Envox d.o.o.

Sheet: /AC power, DC power, AC sync/
File: ac_dc_power.kicad_sch

Title: BB3+ mainboard

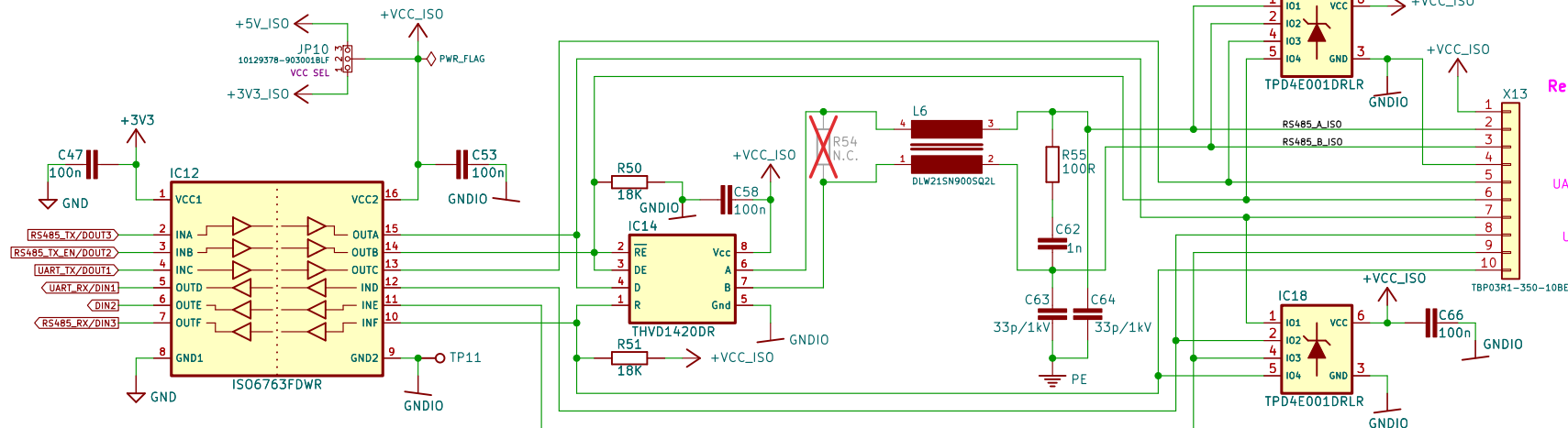
Size: A4 Date: 2026-09-04

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Rev: r1B2

Id: 4/10

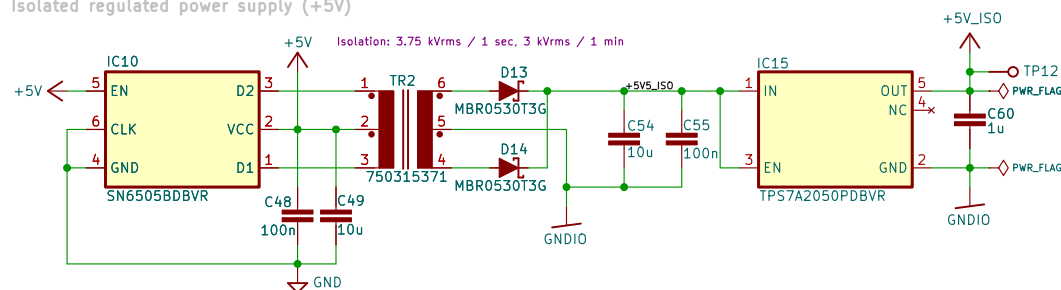
Isolated digital I/O & RS485 transceiver (half-duplex)



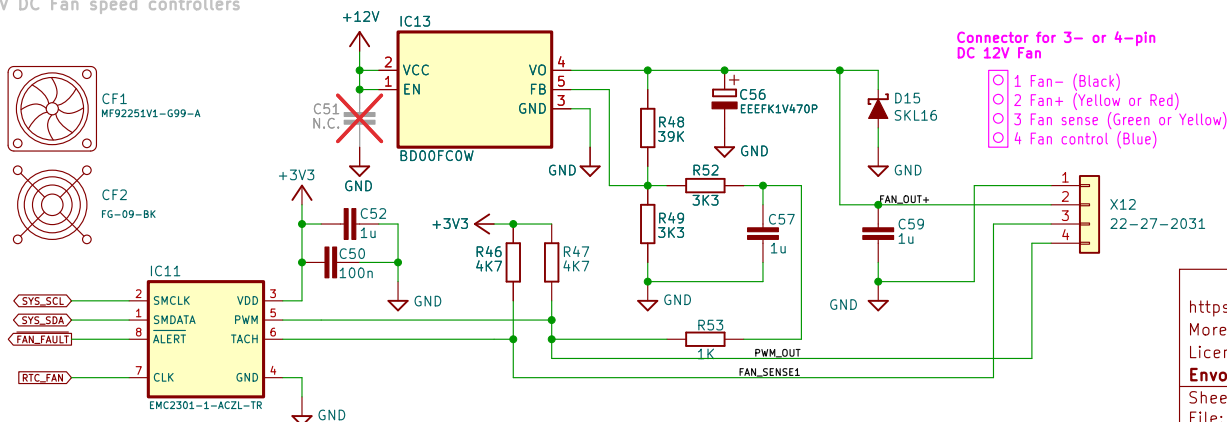
Rear panel I/O connector

- +VCC_ISO 1
- RS485_A_ISO 2
- RS485_A_ISO 3
- GND_ISO 4
- UART_TX/DOUT1 5
- DOUT2 6
- DOUT3 7
- UART_RX/DIN1 8
- DIN2 9
- DIN3 10

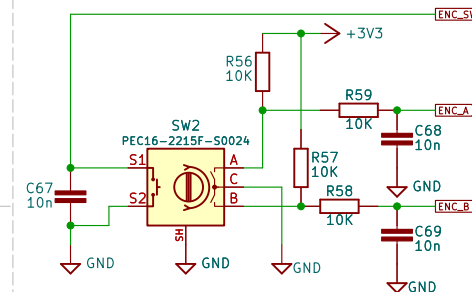
Isolated regulated power supply (+5V)



12V DC Fan speed controllers



Encoder with switch



Encoder shaft length: 15 mm / Flatted

Knob head Ø31mm A1031648 (OKW) + Knob pointer A1101007 (OKW)
Optional: Knob head Ø31mm A3131068 (OKW) + Knob cap A3231007 (OKW)
Knob head Ø24mm A1024648 (OKW) + Knob pointer A1101007 (OKW)

<https://github.com/eez-open/eez-bb3plus>

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Envox d.o.o.

Sheet: /Isolated digital IO, Fan controller, Encoder/

File: digital_io_fan_encoder.kicad_sch

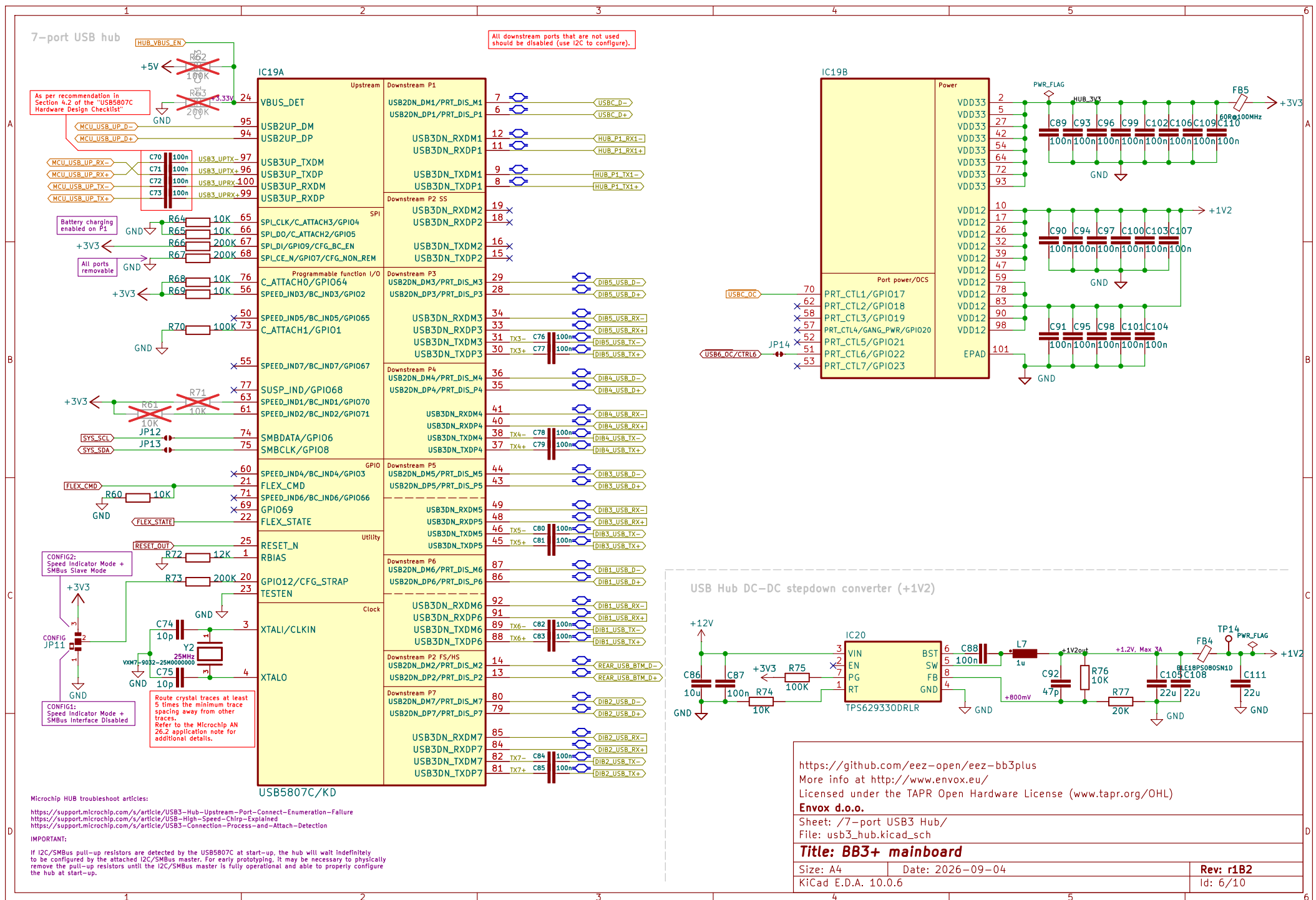
Title: BB3+ mainboard

Size: A4 Date: 2026-09-04

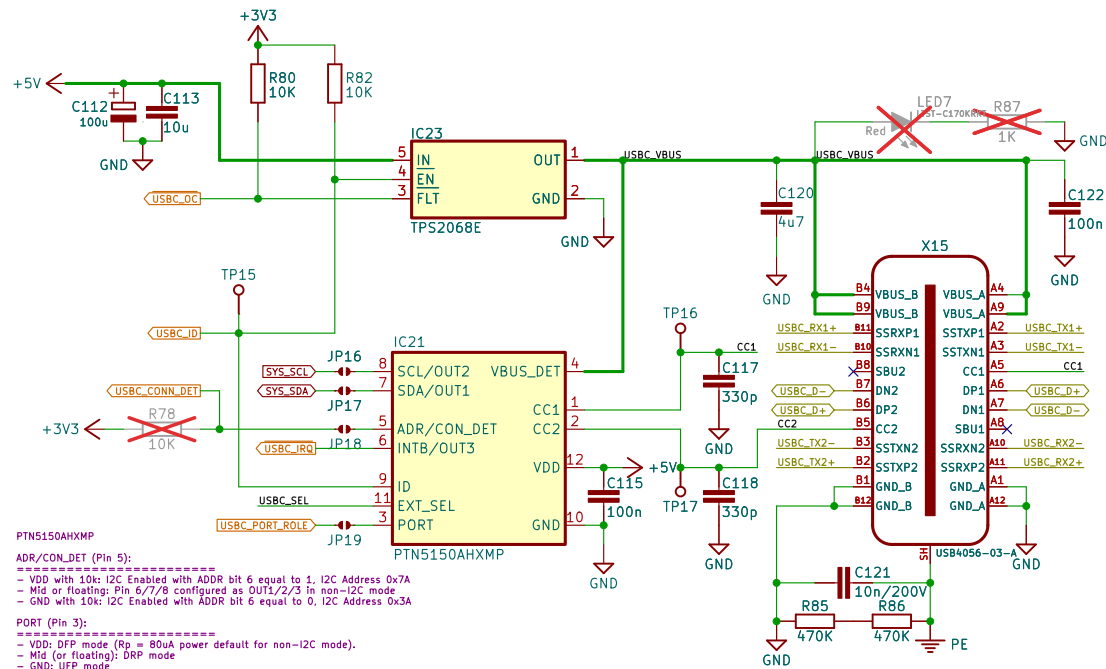
KiCad E.D.A. 10.0.6

Rev: r1B2

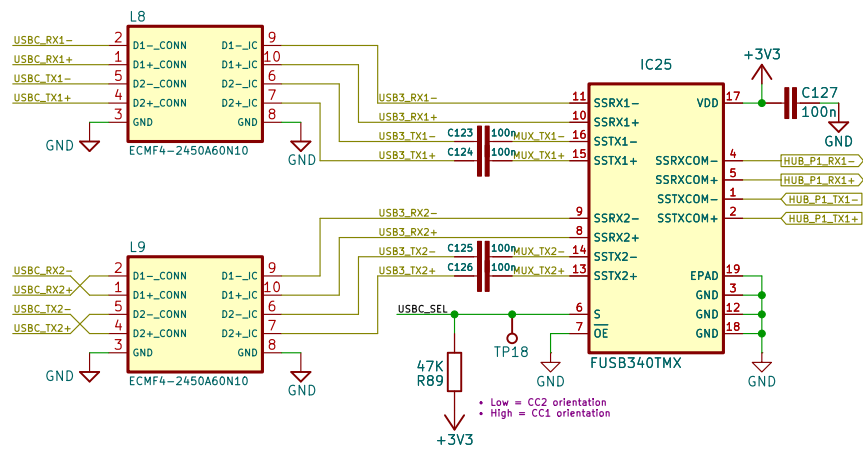
Id: 5/10



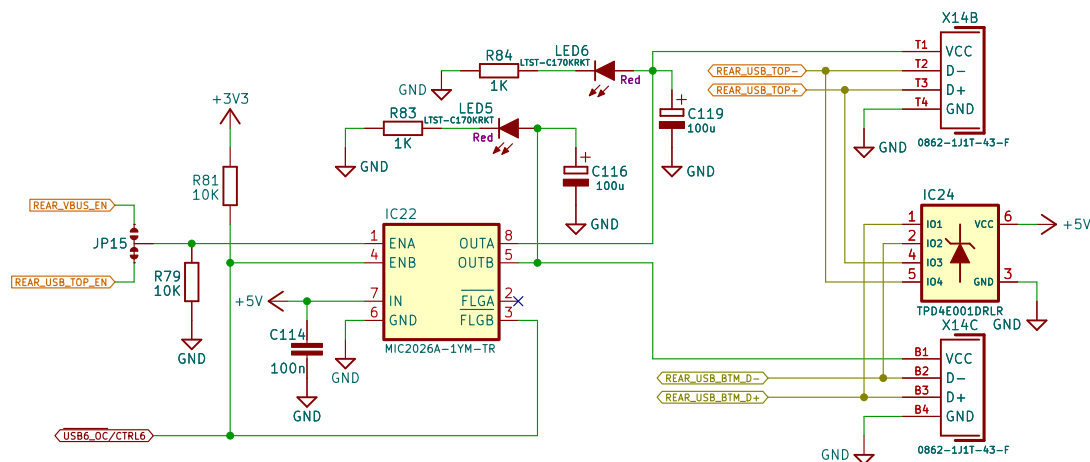
USB-C 3.0 front port (source max 1.5A)



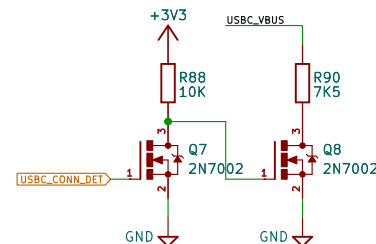
USB-C 3.0 Protection and Mux



USB2.0 Type-A (Rear panel)



VBus Discharge Circuit



<https://github.com/eez-open/eez-bb3plus>

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Envox d.o.o.

Sheet: /Rear USB, USB-C/

File: usb_conn.kicad_sch

Title: BB3+ mainboard

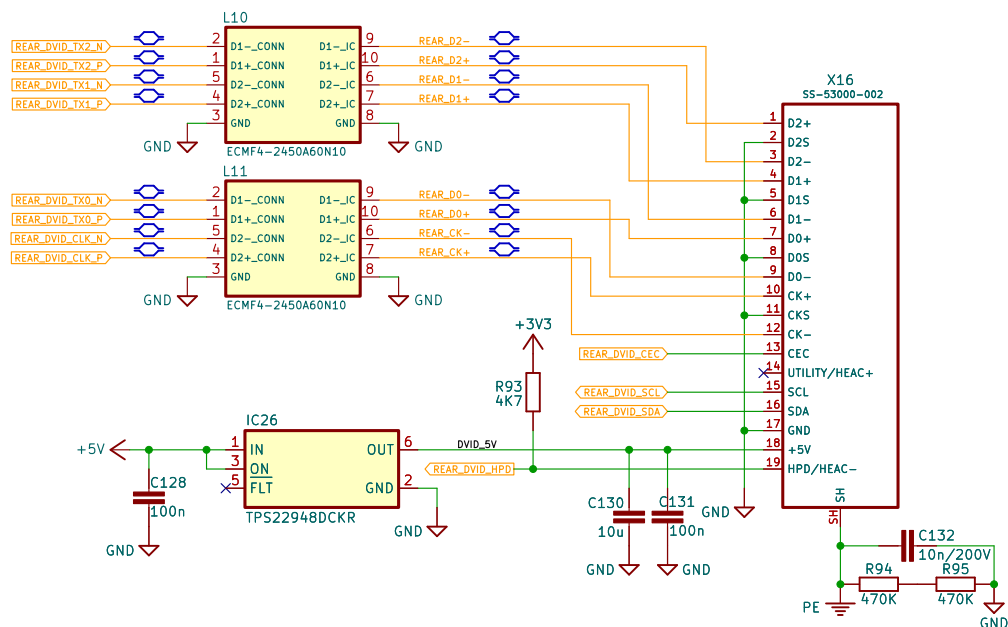
Size: A4 Date: 2026-09-04

KiCad E.D.A. 10.0.6

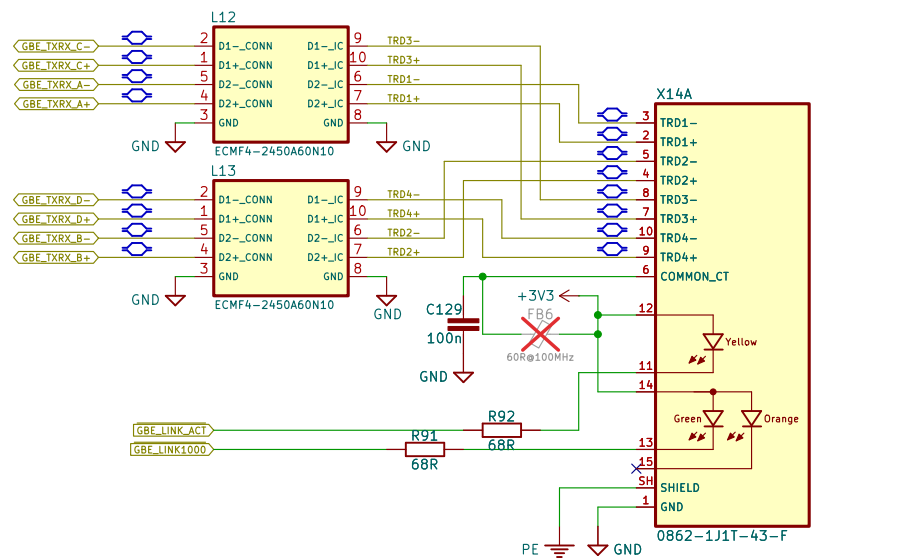
Rev: r1B2

Id: 7/10

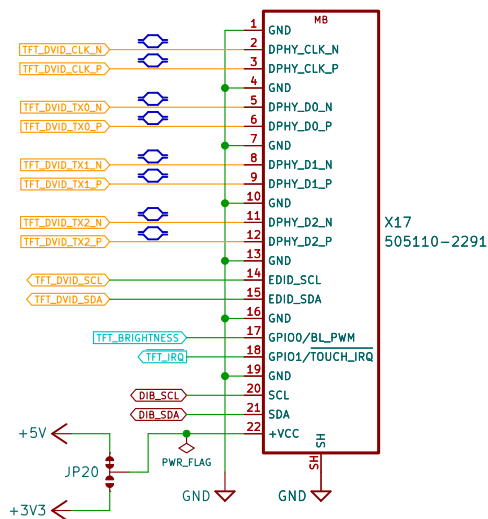
External monitor digital output



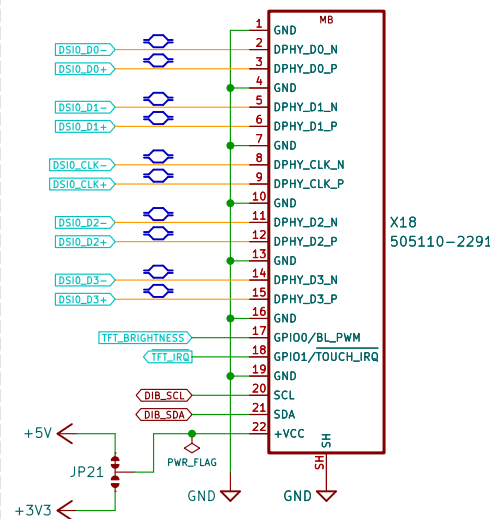
Ethernet RJ45+mag



TFT Touchscreen display DVID output (22-pin FFC)



TFT Touchscreen display DSI output (22-pin FFC)



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Envox d.o.o.

Sheet: /Ethernet, Video outputs/
 File: video_outputs.kicad_sch

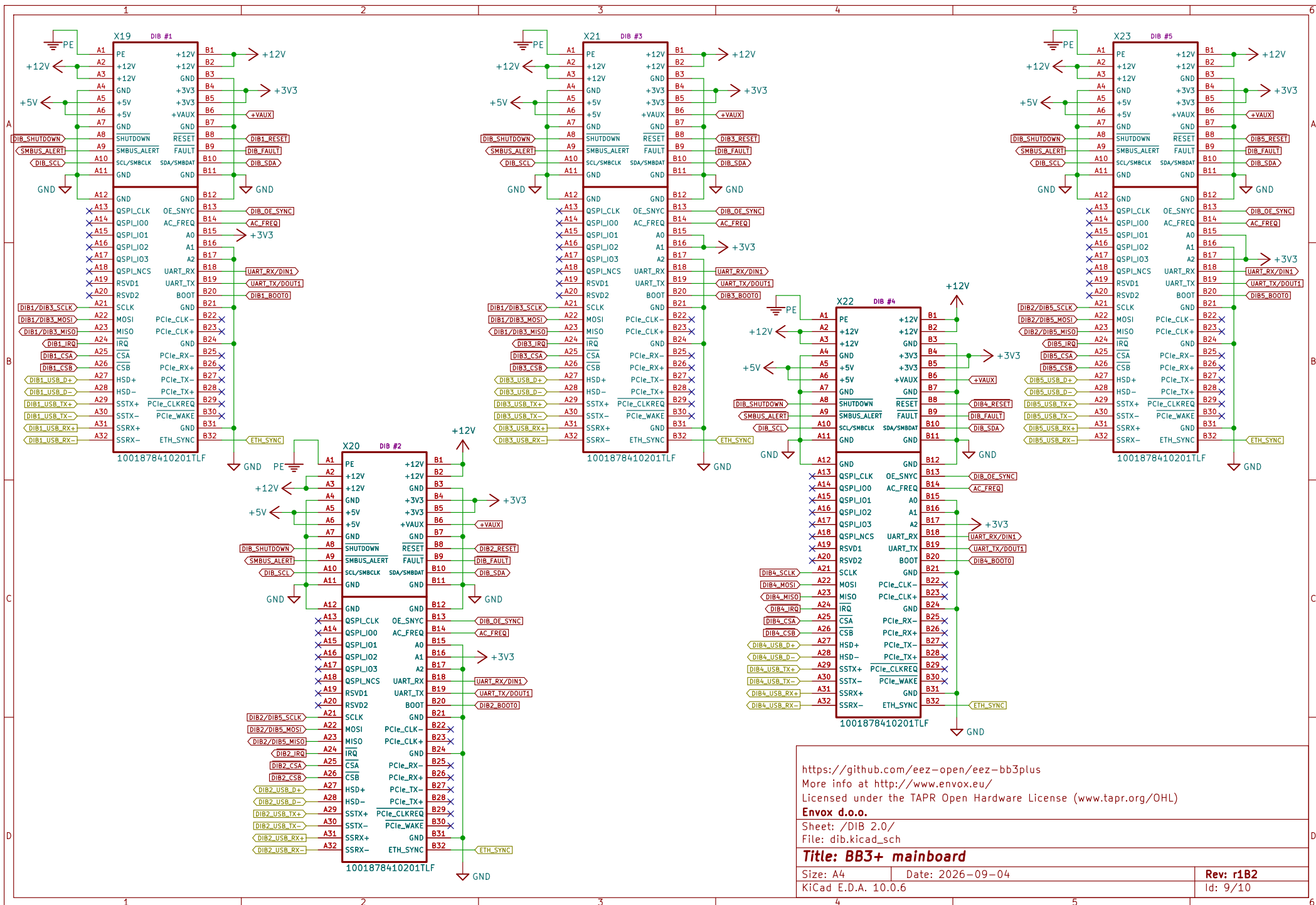
Title: BB3+ mainboard

Size: A4 Date: 2026-09-04

KiCad E.D.A. 10.0.6

Rev: r1B2

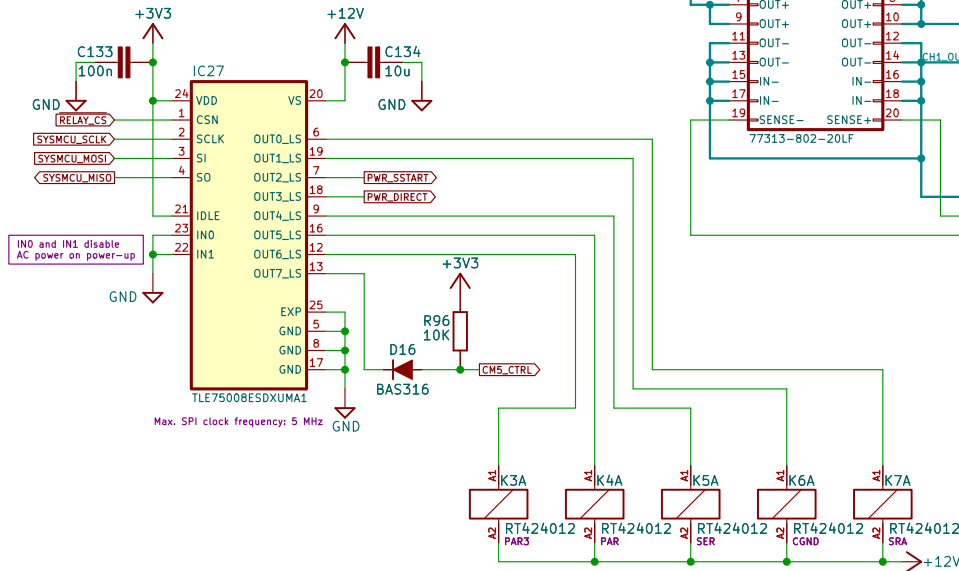
Id: 8/10



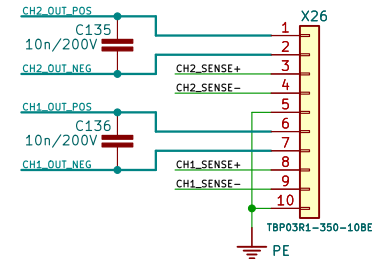
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

PWR+ IN 1	2 PWR+ IN
PWR+ IN 3	4 PWR+ IN
PWR+ IN 5	6 PWR+ OUT
PWR+ OUT 7	8 PWR+ OUT
PWR+ OUT 9	10 PWR+ OUT
PWR- OUT 11	12 PWR- OUT
PWR- OUT 13	14 PWR- OUT
PWR- IN 15	16 PWR- IN
PWR- IN 17	18 PWR- IN
SENSE- 19	20 SENSE+



Rear front power outputs



Rear panel power connector

CH2_OUT+ 1	○
CH2_OUT- 2	○
CH2_SENSE+ 3	○
CH2_SENSE- 4	○
PE 5	○
CH1_OUT+ 6	○
CH1_OUT- 7	○
CH1_SENSE+ 8	○
CH1_SENSE- 9	○
PE 10	○

<https://github.com/eez-open/eez-bb3plus>

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Envox d.o.o.

Sheet: //Power modules coupling/

File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4 Date: 2026-09-04

KiCad E.D.A. 10.0.6

Rev: r1B2

Id: 10/10